

The Ansatz in Differential Equations

A companion document: **Part I** is the theory (what an ansatz is and why it is legitimate), **Part II** is a set of fully worked examples, and **Part III** is a one-page cheat sheet.

Part I — Theory

1. The word

Ansatz is German for “starting point,” “assumption,” or “what you put forward.” In the context of differential equations (DEs) it has a precise technical meaning:

An ansatz is a *form* you propose for the unknown solution — a template containing undetermined parameters — before you have actually solved the equation. You substitute this template back into the DE, and the equation then becomes algebraic (or a simpler equation) that you solve for the unknown parameters.

The crucial point: you are **not** guessing the numerical answer. You are guessing the *shape* of the answer, on the basis of some reason, and letting the equation fill in the details.

2. Why a guess is even legitimate

A DE constrains a function, but a DE *family* has infinitely many solutions, so the equation alone rarely pins down a single one. When you add a boundary/initial condition, the solution space shrinks to a finite-dimensional one. The ansatz is your way of *parameterizing* that (hopefully) finite-dimensional space: you write down a function with the right number of free knobs, and the DE tells you what those knobs must be.

Two things make this work in practice:

- **Symmetry.** If the equation and its domain share a symmetry (translation-invariant coefficients, a separable geometry, a periodic forcing), the solution often inherits it, and that symmetry tells you the shape.
- **Structure of the operator.** Linear operators with constant coefficients act on exponentials by multiplying them — exponentials are their eigenfunctions. That single fact drives the most common ansatz of all.

3. The general workflow

Every ansatz method follows the same four-step pattern:

1. **Propose a form** $y(x) = \text{template}(x; \text{parameters})$.
2. **Substitute** it into the differential equation.
3. **Solve** the resulting (simpler) conditions — usually an algebraic equation — for the parameters.
4. **Verify** the result, and check whether the ansatz is *complete* (i.e. it does not silently miss solutions).

The whole method is only as good as the quality of the reason behind step 1. A good ansatz is *informed*, not random.

4. The classic ansätze

(a) The exponential ansatz — constant-coefficient ODEs

For

$$y'' + ay' + by = 0$$

you propose $y = e^{rx}$. Why? Because $\frac{d^n}{dx^n} e^{rx} = r^n e^{rx}$ — differentiation just multiplies the exponential by r , so a linear combination of derivatives *factors out* the e^{rx} :

$$e^{rx}(r^2 + ar + b) = 0 \implies r^2 + ar + b = 0$$

The DE has collapsed into the **characteristic (auxiliary) polynomial**. The roots r_1, r_2 give the solution $y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$ (with the usual modifications for repeated or complex roots). The exponential was the right *shape* because the operator has constant coefficients (translation invariance).

(b) The monochromatic plane-wave ansatz — PDEs

For the wave equation you propose

$$\psi = \psi_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

The reasoning: the medium is homogeneous (in space) and isotropic, with no time-varying sources, so the equation is invariant under spatial translations and time translations. A function that “respects” those symmetries is a **plane wave** — a single spatial frequency $\mathbf{k}$ and a single temporal frequency ω . Substituting turns $\nabla^2 \rightarrow -k^2$ and $\partial_t^2 \rightarrow -\omega^2$, and the PDE collapses to the algebraic **dispersion relation** (e.g. $\omega = c |\mathbf{k}|$).

This is the single most important ansatz in physics: it converts differential equations into algebra, and the algebraic part is where the real physics (the dispersion relation) lives.

(c) Separation of variables

For a PDE on a domain with separable geometry (a rectangle, a sphere), you propose the solution *factors*:

$$u(x, t) = X(x) T(t)$$

Substitution gives $\frac{T'}{c^2 T} = \frac{X''}{X} = -\lambda$, where the left side depends only on t and the right only on x — they can be equal only if both equal a **separation constant** λ . One PDE becomes two ODEs. The ansatz is justified by the geometry of the domain; the constants λ are then fixed by the boundary conditions (this is where eigenvalues enter).

(d) Trial solution by undetermined coefficients — inhomogeneous ODEs

For $y'' + y = \cos x$ you *know* the homogeneous part ($e^{\pm ix}$) and need a particular solution. You propose a trial of the *same form as the forcing*, $y_p = A \cos x + B \sin x$, and solve for A, B . If the forcing is e^{kx} you try Ae^{kx} ; if it is a polynomial you try a polynomial of the same degree. The rule: **the particular solution has the same functional form as the source term** (for constant-coefficient operators).

(e) Power series / Frobenius

When you cannot see a closed-form shape, you ansatz the solution as a series:

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n$$

Substitution turns the DE into a **recurrence relation** linking a_{n+1} to earlier coefficients. The Frobenius variant, $y = (x - x_0)^r \sum a_n (x - x_0)^n$, handles singular points. The “parameters” are now an infinite sequence, but the recurrence fixes them all from a few initial ones.

(f) Perturbation ansatz

When the equation contains a small parameter ε (e.g. a weak nonlinearity), you ansatz the solution as an *expansion in powers of ε* :

$$y = y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \dots$$

Substitution and collecting order by order gives a hierarchy of easier problems (solve for y_0 , then y_1 , etc.). The ansatz is justified by the existence of a small parameter and a known zeroth-order solution.

(g) Trial-function (variational) ansätze

For eigenvalue or boundary-value problems you cannot solve exactly, you propose a *finite family* of plausible functions $y(x; \mathbf{a})$ with a few parameters and minimize an energy functional, or project the residual onto a basis (Galerkin / Rayleigh–Ritz). The ansatz is a *basis of trial functions*; the DE is satisfied only approximately, but the error can be made systematically small.

5. What an "ansatz" is *not*

- **Not a blind guess.** A pure guess at the numerical answer is not an ansatz. The ansatz is a *structured template*, and the structure must be justified by symmetry, the operator's properties, or a known small parameter.
- **Not always complete.** This is the key caveat. An ansatz can be *too narrow* and miss solutions. Two famous examples:
 - **Resonance in undetermined coefficients.** If the forcing $\cos x$ matches the homogeneous solution, the naive trial $A \cos x + B \sin x$ is already a homogeneous solution and gives no new information — you must *multiply by x* (the "resonant" ansatz). A good method includes a rule for when to modify the trial.
 - **Exponential ansatz at repeated/complex roots.** The raw e^{rx} template only gives *distinct* exponentials; repeated roots require the xe^{rx} term. You must check the ansatz spans the full solution space (dimension = order of the DE).
- **Not a proof of existence.** Finding *one* solution via an ansatz does not by itself prove you have found *all* of them. Completeness is a separate check.

6. The discipline of using an ansatz well

A rigorous ansatz method always closes with a check. The checklist:

1. **Justification** — why this shape? (symmetry, operator structure, small parameter, domain geometry.)
2. **Sufficiency of parameters** — does the template have as many free parameters as the DE's order / boundary conditions require? If not, it is under-parameterized and will miss solutions.
3. **Substitution & solve** — the DE must be satisfied identically; solve the resulting conditions.
4. **Completeness** — confirm the resulting family has the expected dimension; if the operator is linear, confirm you can superpose; check the special cases (resonance, repeated roots, singular points) where the basic template fails.
5. **Boundary/initial conditions** — the ansatz usually leaves free constants; these are what the conditions fix.

Part II — Worked examples

Each example follows the four-step workflow (propose → substitute → solve → verify) and closes with the completeness check.

Example 1 — Exponential ansatz on a homogeneous ODE (complex roots)

Problem. Solve

$$y'' + 4y' + 5y = 0.$$

Step 1 — Propose. Constant coefficients $\rightarrow$ translation invariance $\rightarrow$ exponential ansatz

$$y = e^{rx}.$$

Step 2 — Substitute. Since $y' = re^{rx}$ and $y'' = r^2e^{rx}$,

$$e^{rx}(r^2 + 4r + 5) = 0 \implies r^2 + 4r + 5 = 0.$$

Step 3 — Solve.

$$r = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 5}}{2} = \frac{-4 \pm \sqrt{-4}}{2} = -2 \pm i.$$

Complex conjugate roots $r = \alpha \pm i\beta$ with $\alpha = -2$, $\beta = 1$ give the two independent *real* solutions $e^{\alpha x} \cos(\beta x)$ and $e^{\alpha x} \sin(\beta x)$:

$$y(x) = e^{-2x}(C_1 \cos x + C_2 \sin x).$$

Step 4 — Verify & completeness. Differentiate: $y' = e^{-2x}[-2(C_1 \cos x + C_2 \sin x) + (-C_1 \sin x + C_2 \cos x)]$, and a direct substitution (or the standard theorem) confirms the DE. The two constants C_1, C_2 match the order of the DE, so the family is complete. Physically: a **damped oscillation** — the real part $\alpha = -2$ gives the exponential decay, the imaginary part $\beta = 1$ the oscillation.

Example 2 — Undetermined coefficients and the resonance pitfall

Problem. Solve

$$y'' + y = \cos x.$$

Step 1 — Homogeneous part. $r^2 + 1 = 0 \Rightarrow r = \pm i$, so

$$y_h = C_1 \cos x + C_2 \sin x.$$

Step 2 — Propose a particular solution. The forcing is $\cos x$, so the naive trial is

$$y_p = A \cos x + B \sin x.$$

But this has the *same form* as y_h — it is already a homogeneous solution, so substituting it gives $0 = \cos x$, a contradiction. This is **resonance**: the forcing frequency coincides with a natural frequency of the homogeneous operator.

Resonant ansatz. Multiply the trial by x :

$$y_p = x(A \cos x + B \sin x) = Ax \cos x + Bx \sin x.$$

Step 3 — Substitute. Compute the derivatives:

$$y'_p = (A + Bx) \cos x + (B - Ax) \sin x$$

$$y''_p = (2B - Ax) \cos x + (-2A - Bx) \sin x$$

Then

$$y_p'' + y_p = (2B - Ax + Ax) \cos x + (-2A - Bx + Bx) \sin x = 2B \cos x - 2A \sin x.$$

Set equal to the forcing $\cos x$ and match coefficients:

$$2B = 1 \Rightarrow B = \frac{1}{2}, \quad -2A = 0 \Rightarrow A = 0.$$

Step 4 — Verify & completeness.

$$y_p = \frac{x}{2} \sin x, \quad y_p' = \frac{1}{2} \sin x + \frac{x}{2} \cos x, \quad y_p'' = \cos x - \frac{x}{2} \sin x,$$

so $y_p'' + y_p = \cos x$. □ General solution:

$$y(x) = C_1 \cos x + C_2 \sin x + \frac{x}{2} \sin x.$$

The extra $\frac{x}{2} \sin x$ term grows without bound — the hallmark of resonance (energy pumped in phase with the natural mode).

Lesson. The naive ansatz *failed* and a completeness/justification check caught it. Always ask: *is my trial already in the homogeneous solution space?*

Example 3 — Plane-wave ansatz on the wave equation (PDE)

Problem. Solve the scalar wave equation in a homogeneous medium

$$\nabla^2 \psi - \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2} = 0.$$

Step 1 — Propose. The equation is invariant under spatial and time translations → plane-wave ansatz

$$\psi(\mathbf{x}, t) = \psi_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}.$$

Step 2 — Substitute. The derivatives act as:

$$\nabla^2 \psi = -|\mathbf{k}|^2 \psi = -k^2 \psi, \quad \frac{\partial^2 \psi}{\partial t^2} = -\omega^2 \psi.$$

So

$$(-k^2) \psi - \frac{1}{c^2} (-\omega^2) \psi = \left(k^2 - \frac{\omega^2}{c^2} \right) \psi = 0.$$

Step 3 — Solve. For a nontrivial $\psi_0 \neq 0$:

$$k^2 = \frac{\omega^2}{c^2} \implies \boxed{\omega = c |\mathbf{k}|}$$

(the **dispersion relation**). The PDE has collapsed to one algebraic equation.

Step 4 — Verify & completeness. Substitution is exact, so every monochromatic plane wave with $\omega = ck$ is a solution. Because the equation is *linear with constant coefficients*, the full general solution is a **superposition** (Fourier integral) of such plane waves:

$$\psi(\mathbf{x}, t) = \int \tilde{\psi}_0(\mathbf{k}) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega(\mathbf{k})t)} d^3 k, \quad \omega(\mathbf{k}) = c |\mathbf{k}|.$$

So the single-mode ansatz is a *basis* for the whole solution space — complete in the sense of a Fourier basis.

Connection to Maxwell. This is precisely the ansatz used in the wave-equation derivations: $\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ gives $\omega = c|\mathbf{k}|$, and the *remaining* Maxwell equations (Faraday, Gauss) then fix the free amplitude and polarization: $\mathbf{B}_0 = \hat{\mathbf{k}} \times \mathbf{E}_0/c$ (transversality, $\mathbf{k} \cdot \mathbf{E}_0 = 0$). The ansatz collapses the differential part to algebra; the other equations fix what the ansatz left open.

Example 4 — Side by side: the ODE characteristic polynomial is a PDE dispersion relation

Examples 1 and 3 look different but are the *same operation*. Both operators have constant coefficients, so the exponential ansatz turns every derivative into a multiplication by a "frequency." The only difference is how many frequencies are present.

Left — the ODE $y'' + 4y' + 5y = 0$, ansatz $y = e^{rt}$ (one frequency r):

derivative	result
y	e^{rt}
y'	$r e^{rt}$
y''	$r^2 e^{rt}$

Factoring out e^{rt} :

$$r^2 + 4r + 5 = 0 \quad (\text{characteristic polynomial in one frequency } r).$$

Right — the 1D wave equation $\frac{1}{c^2} u_{tt} - u_{xx} = 0$, ansatz $u = e^{i(kx - \omega t)}$ (two frequencies k, ω):

derivative	result
u	$e^{i(kx - \omega t)}$
u_t	$-i\omega u$
u_{tt}	$-\omega^2 u$
u_x	$ik u$
u_{xx}	$-k^2 u$

Factoring out the exponential:

$$-\frac{\omega^2}{c^2} + k^2 = 0 \implies \omega^2 = c^2 k^2 \quad (\text{dispersion relation in two frequencies } k, \omega).$$

The unifying statement. Substituting the exponential ansatz always yields an algebraic equation in the frequencies — a **dispersion relation**. The ODE characteristic polynomial is simply the special case with a single frequency.

To see it directly, rewrite the ODE with the physics-style convention $y = e^{-i\omega t}$ (so $y' \rightarrow -i\omega$, $y'' \rightarrow -\omega^2$):

$$(-\omega^2) + 4(-i\omega) + 5 = 0 \implies \omega^2 + 4i\omega - 5 = 0.$$

This is a dispersion relation in ω — the same object as the ODE's characteristic equation, just written in frequency language. Its roots are $\omega = -2 \pm i$ (the same r from Example 1 with $r = -i\omega$), giving the damped-oscillation behaviour.

Take-away. "Characteristic polynomial" and "dispersion relation" are one idea: the algebraic condition you get when a constant-coefficient operator acts on an exponential. An ODE has one frequency; a PDE has as many as it has independent variables.

Part III — Cheat sheet

Ansatz	Used for	Why it works	What substitution gives
e^{rx}	constant-coeff. homogeneous ODE	translation invariance	characteristic polynomial in r
$e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$	wave / PDE in homogeneous medium	space+time translation invariance	dispersion relation $\omega(\mathbf{k})$
$X(x)T(t)$	PDE on separable geometry	domain geometry	two ODEs + separation constant
same form as forcing	constant-coeff. inhom. ODE	operator maps the forcing form to itself	algebraic system for coefficients
$\sum a_n(x-x_0)^n$	unknown closed form / singular points	analyticity	recurrence for a_n
$\sum_{n\geq 0} \varepsilon^n y_n$	small parameter ε	regular perturbation	hierarchy of easier DEs
finite trial basis	eigenvalue / BVP, inexact	variational / projection	algebraic eigenvalue problem

The four-step workflow (always):

1. **Propose** a form with undetermined parameters.
2. **Substitute** into the DE.
3. **Solve** the resulting algebraic / simpler conditions.
4. **Verify** and check **completeness** (dimension, resonance, repeated roots, singular points).

The one rule that prevents most mistakes: before substituting, ask “*is my trial function already in the homogeneous solution space?*” If yes, you are at resonance and must modify the trial (typically multiply by x or raise the polynomial degree).

The unifying idea: for constant-coefficient operators, the exponential ansatz converts the DE into an algebraic **dispersion relation** in the frequencies — an ODE characteristic polynomial is that relation with a single frequency.